Cloud Computing

Assignment-Week 11

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **10** **Total mark: 10 X 1 = 10**

QUESTION 1:

Which capability is the defining feature of Dew computing compared to conventional cloud computing?

A) It requires a permanent high-bandwidth connection to the cloud

B) It provides local independence, letting a device store and process data offline and sync with the cloud when connectivity returns

C) It eliminates the need for any cloud component

D) It performs only data storage and never computation

Answer: B

Solution: Dew computing's defining feature is local independence with later synchronization — devices keep working offline and reconcile with the cloud once reconnected. It complements rather than replaces the cloud, and supports both storage and computation.

QUESTION 2:

A weather-monitoring device installed in a remote mountain region must continue recording and processing measurements for several hours even when the Internet connection is unavailable. Once connectivity is restored, the accumulated data is synchronized with a cloud server. Which computing paradigm best matches this requirement?

A) Edge computing

B) Fog computing

C) Dew computing

D) Centralized cloud computing

Correct Answer: C

Detailed Solution: Dew computing emphasizes autonomous operation at or near end devices. A dew-enabled device can continue functioning during network outages and synchronize its locally generated data with the cloud when connectivity becomes available.

QUESTION 3:

Two users modify the same record independently while their devices are offline. When both devices reconnect, the cloud detects that the two versions conflict. Which mechanism is most directly required to handle this situation?

A) CPU scheduling

B) Conflict detection and resolution during synchronization

C) Container image creation

D) Load balancing

Correct Answer: B
Detailed Solution: Independent offline updates can create conflicting versions of the same data. During synchronization, the system must detect these conflicts and apply an appropriate resolution policy to restore a consistent state.

QUESTION 4:

Allowing Dew devices to operate offline and modify data independently most directly introduces which challenge?

A) Increased CPU virtualization overhead

B) The need to detect and resolve data conflicts when devices reconnect and synchronize

C) Permanent loss of all local data

D) Inability to ever reach the cloud

Answer: B

Detailed Solution: Independent offline modification lets copies of the data diverge, so the main challenge is detecting and resolving conflicts during synchronization to restore consistency — the autonomy-vs-consistency trade-off.

QUESTION 5:

A serverless function has not been invoked for a long period. When the next request arrives, the platform requires additional time to initialize the execution environment.

This delay is known as:

A) Warm start
B) Cold start
C) Cache miss
D) Bootstrapping failure

Correct Answer: B

Detailed Solution: A **cold start** occurs when the serverless platform needs to create or initialize a new execution environment before processing a request. This initialization introduces additional latency.

QUESTION 6:

Why are serverless functions typically designed to be stateless?

A) To reduce electricity consumption
B) Because local storage is unlimited
C) To allow automatic scaling and parallel execution without dependency conflicts
D) To avoid using APIs

Correct Answer: C

Detailed Solution: Statelessness ensures each function invocation is independent, allowing the platform to scale horizontally and run multiple instances in parallel without coordination issues.

QUESTION 7:

A team wants to run a batch job that takes about 45 minutes to complete in a single AWS Lambda

invocation. Why is this NOT feasible?

A) Lambda has no internet access

B) Lambda enforces a maximum execution timeout of 15 minutes per invocation

C) Lambda cannot process batch data

D) Lambda functions must always be stateful

Answer: B

Solution: A single AWS Lambda invocation is capped at 15 minutes (900 seconds), so a 45-minute job cannot finish in one invocation. Such workloads must be split into smaller units or run on a service without this limit (e.g., containers, AWS Batch, or EC2).

QUESTION 8:

A serverless application receives 1,000 independent requests simultaneously. The provider can create multiple function instances to process these requests concurrently.

Which property of serverless computing makes this possible?

A) Stateful execution
B) Horizontal scalability of independent function invocations
C) Fixed allocation of one VM per user
D) Permanent local storage

Correct Answer: B

Detailed Solution:
Serverless platforms can automatically create multiple function instances as demand increases. Stateless or independently executable functions are especially suitable for such parallel execution and horizontal scaling.

QUESTION 9:

A research group wants to execute a scientific computation that requires **40 minutes of uninterrupted execution** for each task. The chosen FaaS platform imposes a maximum execution

time of **15 minutes per invocation**.

What is the primary problem with using this platform directly?

A) Functions cannot communicate with databases.

B) The execution time exceeds the platform's maximum invocation duration.

C) FaaS cannot process numerical calculations.

D) Serverless functions cannot execute in parallel.

Correct Answer: B

Detailed Solution: A FaaS platform imposes a maximum execution duration for each invocation. A computation requiring 40 minutes cannot be completed in a single invocation when the platform limit is 15 minutes.

QUESTION 10:

A company needs a managed cloud platform specifically designed for **large-scale analytical queries over structured datasets**, using distributed processing rather than relying on a single database server. Which service is most appropriate?

A. Amazon Redshift
B. Amazon EC2
C. AWS Lambda
D. Amazon Route 53

Correct Answer: A

Detailed Solution:
Amazon Redshift is a managed cloud data warehouse designed for large-scale analytical workloads. It uses distributed query processing and column-oriented techniques to efficiently analyze large datasets.